

0. Definitions

$$h[k] = h_d[k] - \delta h[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_{hd}[k] := \left. \frac{da_h}{dh} \right|_{h_d[k]}, \quad a_{det}[k] := a_h(h_d[k]), \quad a_r[k] := a_h[k] - a_{det}[k]$$

$$\delta h[k+1] = \lambda_c \delta h[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k]$$

$$s[k] := a_{sig}[k] - \hat{a}_{det}[k] \quad (a_{sig} = \text{source signal, §4; } \hat{a}_{det} = \text{maintained reference, §2})$$

1. True — pointwise, any trajectory

$$\begin{aligned} a_h[k+1] &= a_h(h_d[k] + \Delta h_d[k] - \lambda_c \delta h[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k]) \\ &\approx a_h[k] + a'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) a'_{hd}[k] \delta h[k] + a'_{hd}[k] F_{dh}[k] e_{a_h}[k] + a'_{hd}[k] \varepsilon_h[k] \end{aligned}$$

$$\boxed{a'_{hd}[k+1] = a'_h(h_d[k] + \Delta h_d[k]) \approx a'_{hd}[k]}$$

$$\boxed{a_r[k] = a_h[k] - a_{det}[k] = -a'_{hd}[k] \delta h[k] \quad (\text{pointwise; holds for every } h_d(t))}$$

True-signal statistic (sign prior $a'_{hd} > 0$):

$$a'_{hdm}[k] := +\sqrt{\text{Var}(a_r[k]) / \text{Var}(\delta h[k])} = a'_{hd}[k] \quad a'_{hd}[k+1] = \beta a'_{hd}[k] + (1 - \beta) a'_{hdm}[k]$$

2. Reference $\hat{a}_{det}$ — maintained, no wall model

$$\hat{a}_{det}[k+1] = \hat{a}_{det}[k] + \hat{a}'_{hd}[k] \Delta h_d[k], \quad \hat{a}_{det}[0] = a_{nom} \quad (\text{far field})$$

Reference error $e_{det} := a_{det} - \hat{a}_{det}$ (first order, $a_{det}[k+1] = a_{det}[k] + a'_{hd} \Delta h_d$):

$$e_{det}[k+1] = e_{det}[k] + e_{a'_h}[k] \Delta h_d[k]$$

Constant part of e_{det} : cancels in every variance below. Increment part: driven by $e_{a'_h} \Delta h_d$ only $\Rightarrow$ vanishes as $\hat{a}'_{hd} \rightarrow a'_{hd}$ (self-limiting; quantification pending, P1).

3. Construction — no mean-tracker stage

$$\begin{aligned} s[k] &= a_{sig}[k] - \hat{a}_{det}[k] & \hat{\sigma}_s^2[k+1] &= (1 - a_{cov}) \hat{\sigma}_s^2[k] + a_{cov} s^2[k+1] \\ & & \hat{\sigma}_{\delta h}^2[k+1] &= (1 - a_{cov}) \hat{\sigma}_{\delta h}^2[k] + a_{cov} \delta h_m^2[k+1] \end{aligned}$$

$$\boxed{\hat{a}'_{hdm}[k] := +\sqrt{\hat{\sigma}_s^2[k] / \hat{\sigma}_{\delta h}^2[k]}}$$

The deterministic level is removed by exact subtraction, not by filtering: no $\Delta h_d \approx 0$ assumption enters anywhere below.

4. Source content of s

Measurement source ($a_{sig} = a_{xm}$; signal written by physics, $\kappa = 1$):

$$a_{xm}[k] = a_h[k - d] + n_2[k]$$

$$s[k] = -a'_{hd}[k] \delta h[k - d] + n_2[k] + \underbrace{(a_{det}[k - d] - \hat{a}_{det}[k])}_{e_{det} \text{ (P1)}}$$

$$\boxed{\hat{a}'_{hdm}[k] = a'_{hd}[k - d] + n_{var}[k], \quad E[n_{var}] = 0, \quad \text{Var}(n_{var}) = R_{var} \quad (\kappa = 1)}$$

Estimator source ($a_{sig} = \hat{a}_h$; signal written by the estimator itself):

$$s[k] = \hat{a}_h[k] - \hat{a}_{det}[k] = -\hat{a}'_{hd}[k] \delta \hat{h}_3[k] + (\text{corrections})$$

$$\Rightarrow \quad \hat{a}'_{hdm} \approx \hat{a}'_{hd} \quad (\kappa \approx 0, \text{ ruled out; companion doc §3})$$

5. Esti

$$\boxed{\begin{aligned} \hat{a}_h[k + 1] &= \hat{a}_h[k] + \hat{a}'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] \delta \hat{h}[k] \\ \hat{a}'_{hd}[k + 1] &= \beta \hat{a}'_{hd}[k] + (1 - \beta) \hat{a}'_{hdm}[k] \end{aligned}}$$

6. Error

$$e_{a_h} = a_h - \hat{a}_h, \quad e_{a'_h} = a'_{hd} - \hat{a}'_{hd}, \quad e_h = \delta h - \delta \hat{h}$$

Aligned β -forms subtracted ($a'_{hdm} - \hat{a}'_{hdm} = -n_{var}$ at $\kappa = 1$):

$$e_{a'_h}[k + 1] = \beta e_{a'_h}[k] + (1 - \beta) (a'_{hdm}[k] - \hat{a}'_{hdm}[k]) = \beta e_{a'_h}[k] - (1 - \beta) n_{var}[k]$$

$$\boxed{\begin{aligned} e_{a_h}[k + 1] &= (1 + \hat{a}'_{hd}[k] F_{dh}[k]) e_{a_h}[k] + \Delta h_d[k] e_{a'_h}[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] e_h[k] + \hat{a}'_{hd}[k] \varepsilon_h[k] \\ e_{a'_h}[k + 1] &= \beta e_{a'_h}[k] - (1 - \beta) n_{var}[k] \end{aligned}}$$

7. $\mathbf{F}_e$

$$x_e = [\delta h_1, \delta h_2, \delta h_3, e_{a_h}, e_{a'_h}]^T$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{a}'_{hd}[k] & 1 + \hat{a}'_{hd}[k] F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & \beta \end{bmatrix}$$

$$q[k] = [0, 0, -\varepsilon_h[k], \hat{a}'_{hd}[k] \varepsilon_h[k], -(1 - \beta) n_{var}[k]]^\top$$

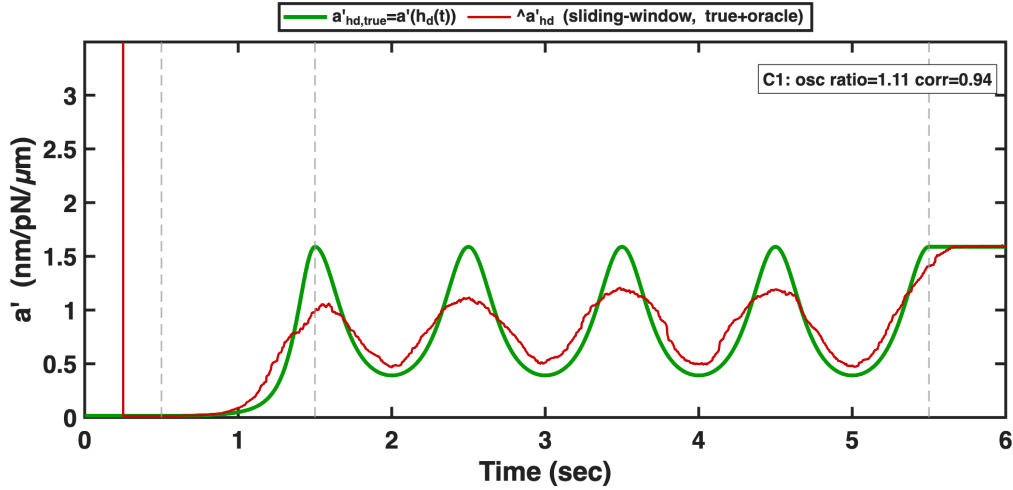
8. Assumptions and pending items

A1: a'_{hd} , $\sigma_{\delta h}^2$ quasi-static in the variance window (slow, K_h -scale) A2: $E[\delta h] = 0$

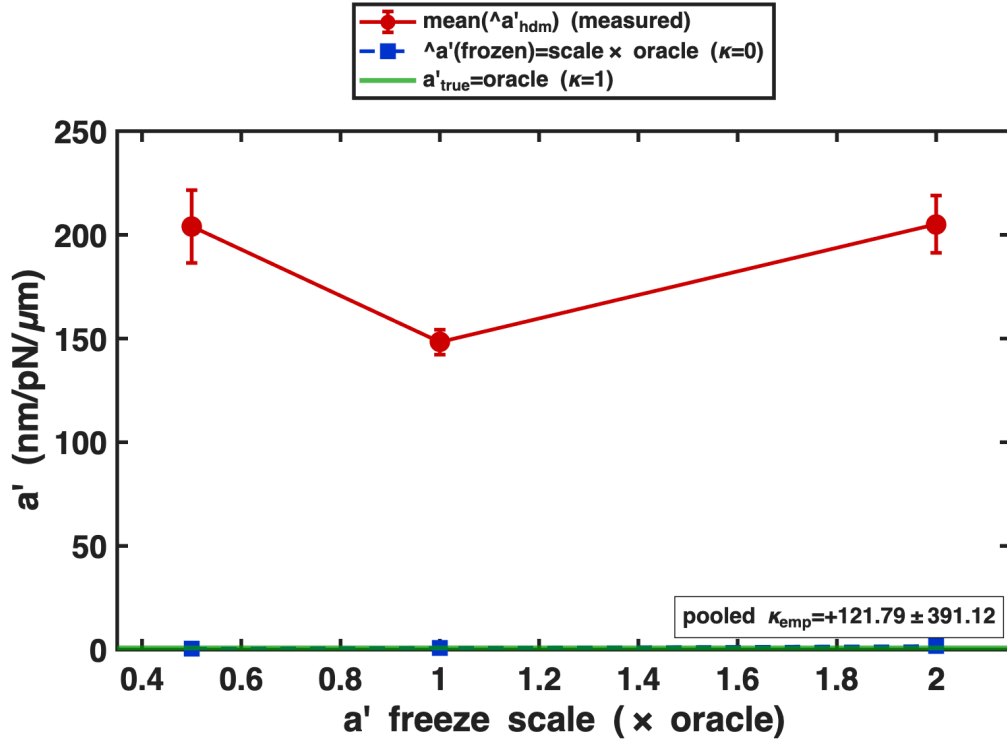
A3: $\delta h \sim \mathcal{N}(0, \sigma_{\delta h}^2)$ A4: $a'_{hd} > 0$ (no $\Delta h_d \approx 0$ assumption)

Pending: P1 e_{det} increment terms (self-limiting, $\propto e_{a'_h} \Delta h_d$); P2 denominator sensor-noise subtraction ($\hat{\sigma}_{\delta h}^2 - C_n \sigma_n^2$); P3 numerator noise statistics (n_2 / R_{22} chain) $\rightarrow R_{var}$; P4 descent-lag mean of δh (A2 violation bookkeeping); P5 β from R_{var} and drift budget; $Q_{55} = (1 - \beta)^2 R_{var} + (\text{drift})$.

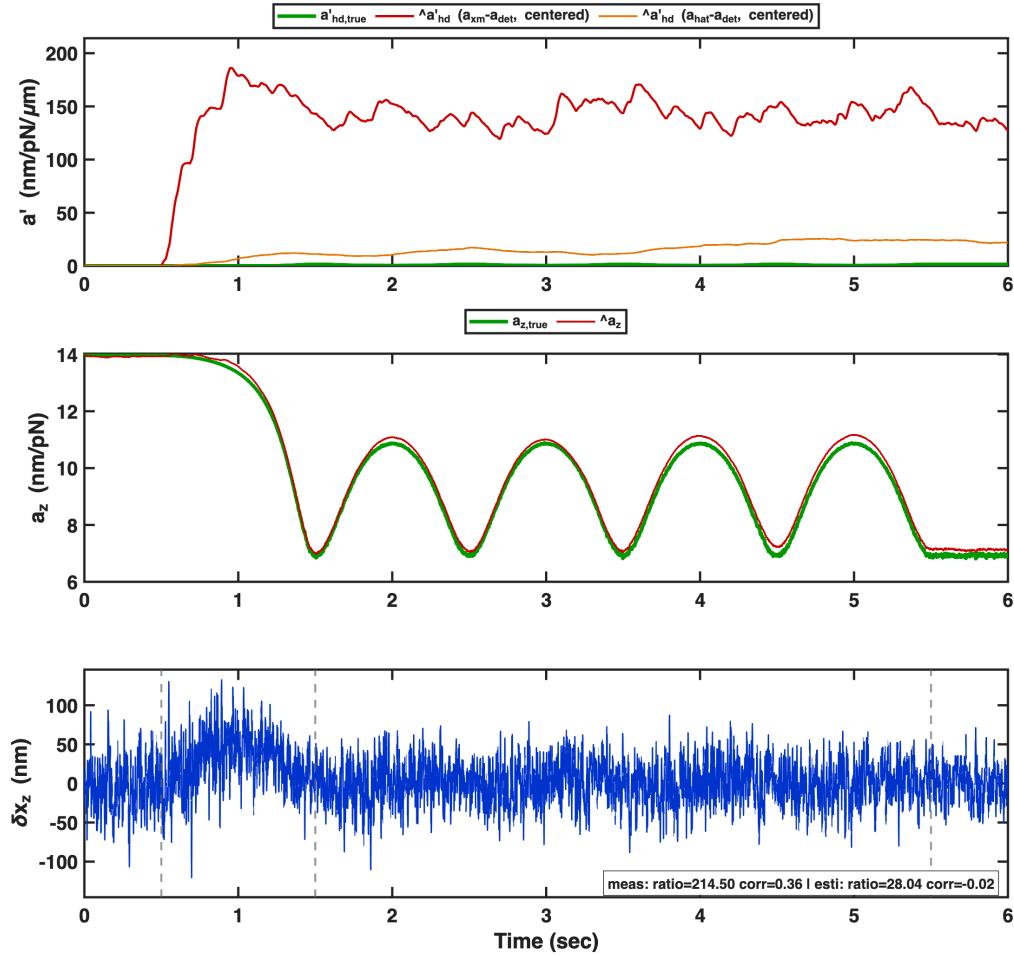
9. Simulation



C1, §1–3 on true signals + oracle reference, 1 Hz full trajectory: osc ratio 1.11 ± 0.03 , corr 0.94 (estimator construction on the same scenario: 13.5 / 0.29) — W1 removed by exact subtraction.

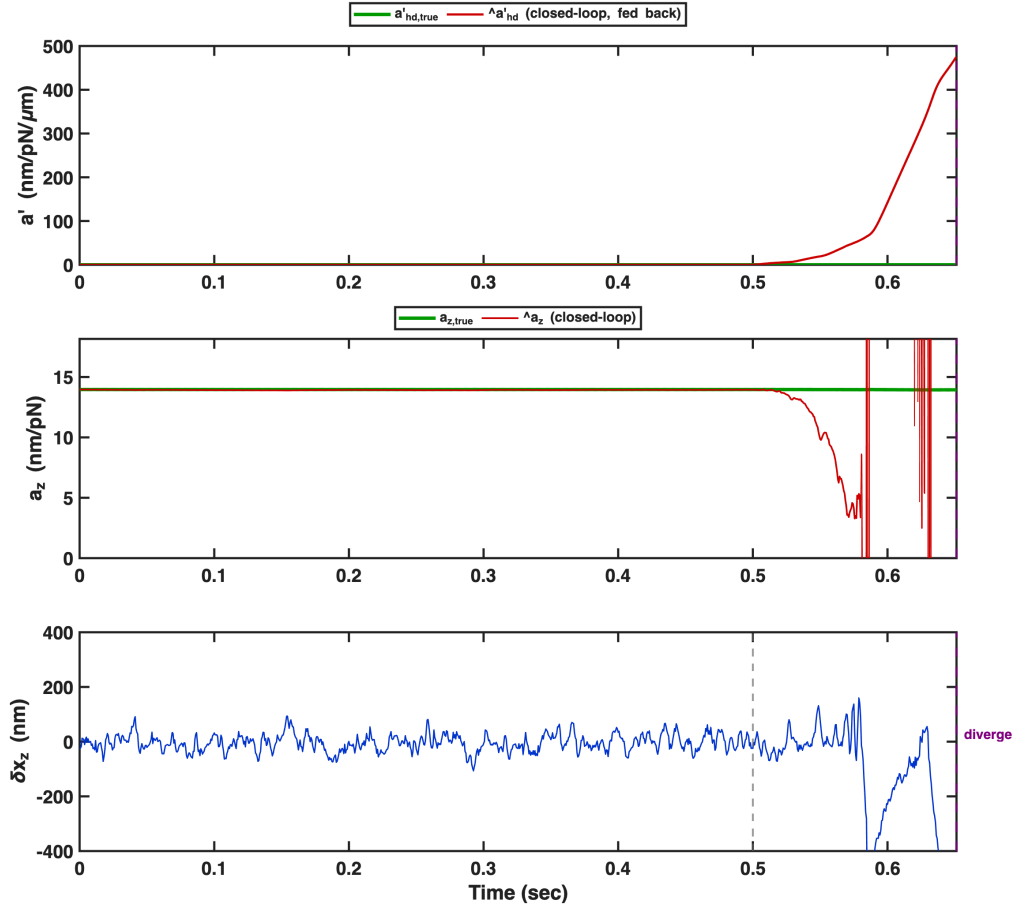


$C3$, measurement source $s = a_{xm} - \hat{a}_{det}[k-d]$, frozen arms: $\hat{a}'_{hdm} \approx 0.15\text{--}0.20$ independent of both $\hat{a}'_{hd}$ and a'_{hd} — $R_{var} \sim 200 \times a'_{hd}{}^2$ noise floor ($P3$), channel buried ($\kappa = 1$ structurally, $SNR \approx 1/200$).



$C4$, shadow EWMA: self-consistent online reference diverges during the descent (always-positive $\hat{a}'_{hdm}$)

$\Rightarrow$ positive feedback in the $\hat{a}_{det}$ recursion, both sources $t \approx 0.75$ s); oracle-decoupled reference stays at the noise floor.



C5, closed loop ($\hat{a}'_{hd}$ fed back to all three a' use points): divergence at $t = 0.651$ s.